Attention-Aware Pseudo-3-D Convolutional Neural Network for Hyperspectral Image Classification

Jianzhe Lin, *Student Member, IEEE*, Lichao Mou, *Student Member, IEEE*, Xiao Xiang Zhu, *Senior Member, IEEE*, Xiangyang Ji, *Senior Member, IEEE*, and Z. Jane Wang, *Fellow, IEEE*

***Abstract*— Convolutional neural networks (CNNs) have been applied for hyperspectral image classification recently. Among this class of deep models, 3-D CNN has been shown to be more effective by learning discriminative features from abundant spectral signatures and spatial contexts in hyperspectral imagery (HSI). However, by simply imposing 3-D CNN to HSI, a large amount of initial information might be lost in this CNN pipeline. The proposed attention-aware pseudo-3-D (AP3D) convolutional network for HSI classification is motivated by two observations. First, each dimension of the 3-D HSI is not equally important, different attention should be paid to different dimensions of the initial HSI image, especially in the first convolution operation. Second, intermediate representations of the 3-D input image at different stages in the 3-D CNN pipeline represent different levels of features and should not be neglected and abandoned. Instead, a 2-D matrix of scores for each feature map should be fed to the final softmax layer. Quantitative and qualitative results demonstrate that the proposed AP3D model outperforms the state-of-the-art HSI classification methods in agricultural and rural/urban data sets: Indian Pines, Pavia University, and Salinas Scene.**

***Index Terms*— Hyperspectral image, salient samples, supervised classification, transfer learning.**

I. Introduction

SUPERVISED hyperspectral image classification has long been investigated for hyperspectral image analysis [1]. Classifying each pixel to a certain land cover type is a primary step for raw hyperspectral imagery (HSI) data analysis. With the advances of deep learning, convolutional neural network (CNN)-based methods have been successfully applied to HSI classification in recent years [2]–[4]. A major advantage of CNN is its extracted spatial feature, which can be integrated with original spectral features learned from balanced local discriminant embedding. Combining spatial and spectral features can help enhance the classification performance [5], [6].

A more direct deep learning model to jointly extract spectral–spatial feature is the 3-D CNN model [7], [8]. This model has shown great potential for HSI classification, as it takes both spectral and spatial correlations of HSI pixels, which are two major characteristics of HSI, into consideration at each convolution layer. Representing the input HSI data with more spatial–spectral information yields promising classification outcomes [7]. However, there is one limitation. This model seems to compress the original HSI data more severely at each convolution layer. Information loss is a major concern for this model [9].

Pseudo-convolutional 3-D (C3D) network has tried to solve such a problem and at the same time has compressed the model size a lot [9]–[11]. The size of a widely acknowledged 11 layers C3D model reaches 321 MB, which is much larger than a 235-MB 2-D Resnet with 152 layers. The pseudo-C3D network would like to split the original C3D model into two parts, one $3 \times 1 \times 1$ convolutional layer and one $1 \times 3 \times 3$ convolutional layer, and these two make up a block. In the block, the two layers would be combined in a parallel or cascaded mode, and by connecting the two layers with different modes, three types of P3D blocks are defined, which are P3D-A, P3D-B, and P3D-C.

However, although the P3D is assumed to be effective, limitation exists when applying this P3D model to HSI classification. First, as different from video data, for HSI, the three dimensions should be of equal importance, and we think the tensor decomposition can be further exploited. The three dimensions can be all separated, creating three 2-D CNN layers: one $3 \times 3 \times 1$, one $1 \times 3 \times 3$, and one $3 \times 1 \times 3$ convolutional layer. These three convolutional layers can describe the original HSI with more details, especially when splitting the first 3-D convolutional layer in the network pipeline which is connected with the original HSI.

However, there are still weak points for such network setting. First, the three separated 2-D convolution layers should not be treated equally, especially for HSI, the importance of spatial and spectral information should not be the same. Second, the intermediate representations of the input image at different stages in the P3D CNN pipeline are ignored, and the middle blocks with different P3D modules almost do not influence the final output deep feature [12]. The middle P3D blocks with different types cannot effectively influence the

Manuscript received September 20, 2020; accepted November 2, 2020. Date of publication February 4, 2021; date of current version August 30, 2021. This work was supported by The National Natural Science Foundation of China under Grants 61620106005. *(Corresponding author: Xiangyang Ji.)*

Jianzhe Lin and Z. Jane Wang are with the Department of Electrical and Computer Engineering, The University of British Columbia, Vancouver, BC V6T 1Z4, Canada.

Lichao Mou and Xiao Xiang Zhu are with the Signal Processing in Earth Observation, Technical University of Munich, 80333 Munich, Germany, and also with the Remote Sensing Technology, German Aerospace Center, 82234 Wessling, Germany.

Xiangyang Ji is with the Department of Electrical and Engineering, Tsinghua University, Beijing 100084, China (e-mail: xyji@tsinghua.edu.cn).

Digital Object Identifier 10.1109/TGRS.2020.3038212

1558-0644 © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

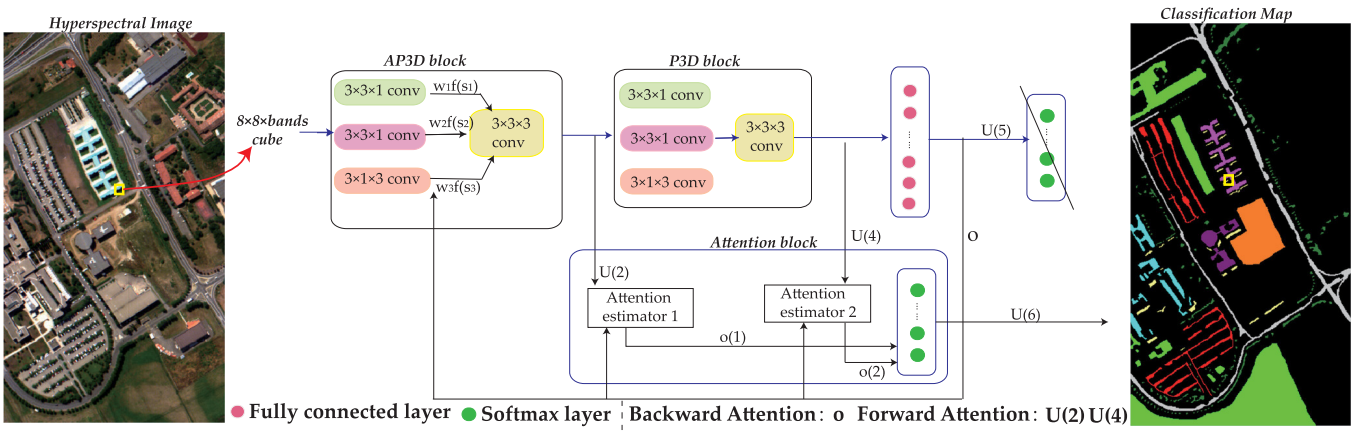


Fig. 1. Flowchart of the proposed AP3D framework for hyperspectral image classification. The input HSI is first divided into 8 × 8 × bands patches, which are fed into the neural network and flow through the P3D blocks. Then, the forward and backward attention learning is conducted iteratively until the maximum number of steps. Finally, the classification map is obtained.

final output feature, as well as the classification performance. Based on such motivation, we propose the attention-aware pseudo-3-D (AP3D) model for HSI classification. The major contributions are claimed as follows. A detailed AP3D pipeline can be found in Fig. 1.

1) *Pseudo-3-D Feature:* We propose to decompose the 3-D convolutional layers into three separate dimensions, this new decoupled feature is named pseudo-3-D feature. Each 3-D layers would be replaced by one 3 × 1 × 3, one 3 × 3 × 1, and one 1 × 3 × 3 convolutional layer, in order to describe each dimension of HSI more efficiently. This setting is especially used for the first 3-D layers in the 3-D CNN pipeline. The three separate 2-D convolutional layers would act as an integrated block and would be added together to input the next block. This 3-D convolutional layer is viewed as new pseudo-3-D (P3D) CNN layer.
2) *Backward Attention Learning:* In order to emphasize on the most precious features of the extracted 3-D HSI feature, we propose such learning process. For each dimension of the extracted 3-D convolutional feature, we adopt the attention model to assign each dimension with different weights, we name this process as backward attention learning. This attention learning mechanism is for the first time proposed. As for the HSI cube, different dimensions are of different importance, the decomposed 3-D convolutional feature should not be treated equally. These weights would be learned by the compatibility between the objective decomposed 3-D feature and the last output feature of the network pipeline and will be assigned to each dimension of the objective 3-D features.
3) *Forward Attention Learning:* To make the best use of most discriminative output features for better final classification, we introduce such learning process. We also propose to assign attention to the last output feature of the neural network pipeline, we name this process forward attention learning. Similar to backward attention learning, the intermediate output representation of input image at different stages (layers) in 3-D the CNN pipeline will be connected with the last deep feature layer, and the compatibility between the middle feature layer and the last deep feature layer will create an attention score. However, opposite to backward attention learning, these weights will further be assigned to the last output feature. The generated optimal output deep feature will be used for the final classification.

The backward and forward attention learning will be conducted iteratively. After these iterations, the proposed framework can get final HSI classification results.

II. Related Work

In this section, two models for HSI classification are reviewed: C3D model and attention model. We also point out a few existing problems in the literature that would be addressed in this article.

A. C3D Neural Network

Convolutional network has been proved to be effective in hierarchically extracting deeper features due to its layer-by-layer structure [13]–[15]. This network has also successfully be applied to hyperspectral image for classification and detection [16], [17]. However, for HSI image due to its spectral and spatial property, barely applying CNN to HSI cannot get the best performance. For spatial–spectral feature extraction, Chen *et al.* [18] at first proposed an HSI classification model based on a deep belief network for which the input is flattened neighbor region. Also, by 2-D CNN, Liang *et al.* extracted the spectral–spatial information by combining sparse representation [19], [20]. Mei *et al.* [21] on the one hand used 2-D CNN to extract spectral features and on the other hand fuse handcrafted spatial features to improve the final performance. Makantasis and Karantzalos [22] integrated spatial–spectral input into CNN (RPCA-CNN) by a random principal component analysis to preprocess pixels in a spatial neighborhood for hyperspectral classification. In [23], the squeeze and excitation method is applied to HSI to explore the spatial–spectral information. Another major method for applying spectral–spatial information for HSI classification should be the superpixel-based method [24]–[26], for which the HSI with different

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

resolution is fully considered. Researchers also apply kernel-based HSI methods [27] to project the original data to a common subspace to exploit the spatial–spectral information of HSI. However, all these frameworks still cannot exploit both spectral and spatial features simultaneously. The two separate feature extracting processes seem not mutually beneficial.

To automatically learn the spatial context of pixels from training samples, using a learnable 3-D kernel should be a reasonable way. Chen *et al.* first used the 3-D convolution to explore the spatial and spectral structure information from the 3-D hyperspectral cube directly [28]. This work is enhanced by Li *et al.* [29] via further fine-tuning the neural network. However, it has been proved that multiscale 3-D deep convolutional neural network (M3D-DCNN) can directly extract both the multiscale spatial feature and the spectral feature for HSI classification and boost the accuracy [30]. By carrying out a final five-layer M3D-DCNN, both 2-D multiscale spatial features and 1-D spectral feature are jointly learned from HSI data in an end-to-end approach. By using different sizes of convolutional kernels simultaneously, another multiscale 3-D CNN network is also proposed [31]. The multiple scales of spectral–spatial features are extracted for a specific pixel by considering its complex spatial context. However, all these 3-D CNN HSI classification works still do not consider the relationship between the spatial and the spectral information, as well as how to assign weights to them when applying them simultaneously for classification.

The proposed framework would split the 3-D CNN into three attention-aware separate dimensions that are named pseudo-C3D. This pseudo-C3D should overcome the problems above.

B. Attention Model

Early work on attention was for video tracking [32], which was motivated by human attention mechanism theories and done via restricted Bolzmann machines [33]. For the regular attention model, it can be used to allocate the available computational resources to the most valuable components of a signal. This model has demonstrated its power in tasks, including image understanding and image captioning [34]. In most time, this model can act as a module to insert into a framework, assigning weights to different parts of images or components of tasks.

The attention model has also been widely used in deep learning [35], [36]. To visualize and interpret the inner characteristic of CNN, attention map, which uses a scalar matrix to represent the relative importance of layer activations at various 2-D spatial locations, is a well-known framework [37]. The "attention" here means the explicit scalar representation of the relative relevance between features being used. A recent typical work achieving this purpose should be the trunk-and-mask attention mechanism based on hourglass modules [38] that are inserted between the intermediate stages of deep residual networks. Another squeeze and excitation block can also be inserted into Resnet by comprising a lightweight gating mechanism, which focuses on enhancing the representational power of the network by modeling channelwise relationships in a computationally efficient manner [39]. The attention-based network has also been recently applied to HSI classification [40], but the limitation of such related work is the method that only consider the spectral attention, but fail to achieve beneficial spatial/spectral attention for HSI classification.

The current trainable attention models for deep learning is mainly designed for query-based task [41]. The existing attention model can be roughly divided into spatial attention models, semantic attention model, and multilayer attention models.

1) *Spatial attention:* The typical tasks using spatial attention include domain adaptation, image captioning, adversarial learning, and so on [42]–[44]. For the domain adaptation work, the attention model is used for handling the domain changes, including the background content, occlusion, and resolution changes. Selectively focusing on the object of interest can effectively improve the transfer performance. For image captions, researchers would use pooling to select the most probably attentive region or average the spatial features with attentive weights. The adversarial feature can also be found by the attention model to increase the adversarial robustness.
2) *Semantic attention:* Semantic attention, paying attention to what things mean, involves using attention-aware resources for object identification and categorization. The most famous work for the semantic attention model should be applied in image captioning. The attention model would be able to find out a semantically important concept or region of interest in an image. For multiple concepts, it is also able to weigh the relative strength of attention paid on, as well as switch attention among concepts dynamically.
3) *Multilayer Attention:* Multilayer attention is quite popular these years. The main idea for this type of attention is in purpose to preserve the middle layer feature representation of input and output it to the softmax layer to help determine the final pattern of inputs. Typical works would be found in [30] and [45]. These works apply the proposed attention mechanism to the popular VGGNet and ResNet, by capturing coarse-to-fine attention maps at multiple levels of CNN pipeline, to bootstrap baseline CNN architectures for the image classification task.

In recent year, this attention is also applied to hyperspectral image classification, which is used for setting up the correlation between the neighboring spectral bands [36], [46]. However, in this work, we would like to introduce the spatial attention in the feature level of the input HSI and the multilayer attention in the pipeline of our CNN classification model. By introducing the P3D layer and attention layers one by one, the comparisons are made, and the effectiveness of each step is verified.

III. Proposed Framework

In this section, we will illustrate the framework in detail. At the very first, we will make clear some conceptions.

The first concept is attention. One approach for interpreting the inner workings of CNNs is the attention map: a scalar matrix representing the relative importance of layer activations at different network locations with respect to the target

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

task [37]. The second concept is local and global features. We use the terms "local features" to refer to features extracted by some layers of CNN whose receptive fields are contiguous proper subsets of the image (like the feature extracted by typical convolutional connected layer), and "global features" to refer to features whose receptive fields are the entire image (like the feature extracted by fully connected layer). By defining a compatibility measurement between local and global features, we can define the attention. We name the process to learn the local features considering its compatibility with global features as backward attention learning. For the global features used to classify the input image, by using a weighted combination of local features, it can highlight the regions of interest while suppressing background clutter of the input image, and we name this global feature learning process as forward attention learning.

A. Backward Attention Learning: P3D Model

The pipeline of the AP3D framework is shown in Fig. 1. In this section, we first define the 3-D convolutions for hyperspectral image representation, which can be decoupled into two 2-D spatial convolutions and one 2-D spectral convolutional filter. This newly designed P3D model is then further devised by leveraging both spatial and spectral filters. By placing this P3D block at different places of the 3-D convolutional network pipeline, both performances and efficiencies are evaluated experimentally. As the local P3D features are weighted based on the global features, this process is backward attention learning.

1) 3-D Convolutions: Given a hyperspectral image with the size of $h \times l \times w$, where h denotes the number of bands and l and w denote the length and width of HSI, respectively, the most natural way to encode such spectral–spatial information should be 3-D convolutions, which simultaneously learn the spatial information as well as the connection between the neighboring bands. To encode these three dimensions' information, the most typical method is to introduce a 3-D kernel denoted as $d \times k \times k$. We further decouple this kernel as one $1 \times k \times k$, one $d \times 1 \times k$ kernel, and one $d \times k \times 1$ kernel. The outputs of these three kernels are weight-based attention mechanisms. We name these kernel-based deep blocks as P3D CNN block and the model as P3D CNN model. These decoupled kernels can encode each dimension of HSI with direct and balanced representations.

2) Attention-Aware HSI 3-D Representation: For the first P3D layer connected with the HSI input, the attention for each dimension is assigned according to its compatibility with the output of the 3-D convolutional network pipeline. As can be found in Fig. 1, we denote a given 3-D convolutional layer as $s \in \{1, 2, \ldots, S\}$ in which S is the total number of layers. If this layer is the proposed P3D convolutional layer, we denote its representation as $F(s) = \{f(s)_1, f(s)_2, f(s)_3\}$ to represent the three dimensions of features. For each $f(s)_i$, the representation of this feature is denoted by $f(s)_i = \{f(s)_i^1, f(s)_i^2, \ldots, f(s)_i^n\}$, in which n is the total number of spatial locations in layer s. The global output feature vector o has the entire image as support and is output by a series of 3-D convolutional and nonlinear layers, after passing through the final two fully connected layers. The compatibility between the feature i at location m $f(s)_i^m$ and o will be calculated by the following equation [12]:

$$c(s)_i^m = \left\langle \hat{f}(s)_i^m, o \right\rangle \tag{1}$$

where $\hat{f}(s)_i$ is the image of $f(s)_i$ under a linear mapping of $f(s)_i^m$ to the dimensionality of o. For $f(s)_i$, the final compatibility should be $C(\hat{f}(s)_i, o) = \{c(s)_i^1, c(s)_i^2, \ldots, c(s)_i^n\}$. For features at each dimension, we can get such $C(\hat{f}(s)_i, o)$. Now, we will further normalize the compatibility at each spatial location for all the one dimension features by a softmax operation, which is formulated as

$$a(s)_i^m = \frac{\exp\left(c(s)_i^m\right)}{\sum_{i=1}^{3} \exp\left(c(s)_i^m\right)} \tag{2}$$

this normalized compatibility score is further formulated as $A(s)_i = \{a(s)_i^1, a(s)_i^2, \ldots, a(s)_i^n\}$ for feature $f(s)_i$. We now will add all the elements in $A(s)_i$ and get the final weight w_i for $f(s)_i$. This weight is further normalized for the three dimensions by

$$\hat{w}_i = \frac{w_i}{\sum_{i=1}^{3} w_i}. \tag{3}$$

Now, for each feature representing one dimension in the P3D block, the feature will be normalized as $\hat{w}_i f(s)_i$, in which $\hat{w}_i$ will multiply with each element in $f(s)_i$. The final output of one P3D block would be

$$I(s) = \hat{w}_1 f(s)_1 + \hat{w}_2 f(s)_2 + \hat{w}_3 f(s)_3. \tag{4}$$

The features will be added at each spatial location and $I(s)$ will finally be gotten, which will pass through a nonlinear relu layer and we can get $U(s)$, which will be input for next C3D block.

B. Forward Attention Learning: Attention-Aware 3-D Network

In this section, we will first introduce the learning process for the global feature and, then, illustrate the whole neural network pipeline in detail.

1) Attention-Aware Global Features: For a random layer $s \in \{1, 2, \ldots, S\}$, we denote its output feature by $U(s)$ as mentioned in the last section. The global features o is the output of a series of convolutional and nonlinear layers. This layer only needs to pass through the final fully connected layer to get the classification score for each class. However, this old architecture does not consider the compatibility between o and middle features $U(s)$.

To introduce the compatibility and get the attention-aware global feature $o(s)$, we have

$$c(s)^m = \langle \hat{U}(s)^m, o \rangle \tag{5}$$

where m is the number of spatial location and $\hat{U}(s)$ is the image of $U(s)$ under a linear mapping of the $U(s)^i$ (here, i represents the spatial location) to the dimensionality of o. Now, the set of compatibility scores can be denoted as $C(\hat{U}(s), o) = \{c(s)^1, c(s)^2, \ldots, c(s)^n\}$, where n is the total spatial locations

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

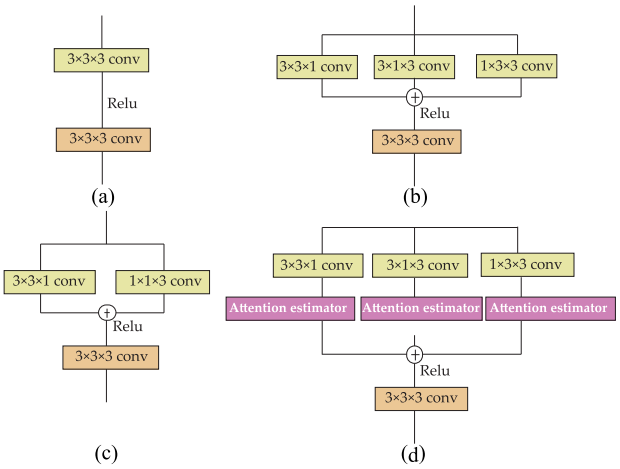


Fig. 2. 3-D block comparisons among C3D, traditional P3D, P3D, and AP3D blocks. (a) C3D block. (b) Traditional pseudo-C3D block. (c) Pseudo-C3D block (for HSI). (d) Attentional pseudo-C3D block.

in o. The compatibility scores are then normalized by a softmax operation

$$b(s)^i = \frac{\exp(c(s)^i)}{\sum_{i=1}^{n} \exp(c(s)^i)}. \tag{6}$$

The normalized compatibility scores $B(s) = \{b(s)^1, b(s)^2, \ldots, b(s)^n\}$ represent the attention. This $B(s)$ is then multiplied with o as the weights to produce single vector $o(s) = \sum_{i=1}^{n} b(s)^i \cdot U(s)^i$ for layer s. Now, $o(s)$ will be input into the last fully connected layer to get the final classification result.

As there are multiple layers, we actually have $O = \{o(1), o(2), \ldots, o(S)\}$, and in correspondence to these outputs, we have final classification probability results as $P = \{p(1), p(2), \ldots, p(S)\}$, in which each $p(s)$ includes t values corresponding to the probabilities of t classes. The last step is to get the average value for P to get the final classification scores for each class.

2) Network Pipeline: In this section, we will introduce the whole neural network pipeline in detail.

We cut down the original HSI to 8×8 small patches as the input. The neural network is mainly set up by several P3D blocks and fully connected layers. For the P3D blocks, as has been indicated in [9], enhancing that the structure diversity with going deep can improve the power of the neural network. We also find that when contacting one Attention-aware P3D/*P3D* with one C3D, the performance would boost. Therefore, we set up the block of this neural network with one attention-aware P3D layer/P3D layer and one C3D layer, and we name this block as attention-aware *P3D block* (AP3D block for short)/P3D block (P3D block for short) for HSI. This block is shown in Fig. 2.

As can be found in Fig. 1, the input image would first pass through one AP3D blocks, one P3D blocks, one fully connected layer, and one softmax layer (the transition layers are neglected here, and the detailed network can be found in Table IV). The output of the fully connected layer would have two outputs. The first is returned to the first AP3D block by backward attention learning to create weights for three AP3D layers; the second is to be weighted by the attentions created by its compatibility with both outputs of AP3D and P3D blocks, and then to be connected with the final softmax layer with t outputs representing t classes, to get the final classification map.

Algorithm 1 Classification for HSI

Input: X
Output: P
function AP3D NETWORK(X)
  while $n < n_e$ **do**
    for $s = 1, 3$ **do**
      if $s = 1$ **then**
        for $i = 1, 2, 3$ **do**
          Calculate $f(s)_i \leftarrow conv_{2d}(X)$
          $U(1)_i \leftarrow$ BACKATT$f(1)_i, U(4)$
        end for
      else
        for $i = 1, 2, 3$ **do**
          Calculate $U(s)_i \leftarrow conv_{2d}U(s-1)$
        end for
        $U(s+1) \leftarrow relu(U(s)^1 + U(s)^2 + U(s)^3)$
        $U(s+1) \leftarrow relu(conv_{3d}(U(s+1))$
      end if
    end for
    $U(5) \leftarrow full_{conv}(U(4))$
    $U(6)_2 \leftarrow$ FORWARDATT$U(5), U(2)$
    $U(6)_4 \leftarrow$ FORWARDATT$U(5), U(4)$
    $o \leftarrow U(6)_2 + U(6)_4$
    $p \leftarrow softmax(o)$
    $backpropagation, n = n + 1$
  end while
end function
function BACKATT(f, U)
  map f to dimension of U to get $\hat{f}$
  Calculate c by Eq. (1)
  Calculate a by Eq. (2)
  $w \leftarrow a(s)_i^1 + a(s)_i^2 + \ldots + a(s)_i^n$
  Calculate $\hat{w}$ by Eq. (3)
  Return $\hat{w} f$
end function
function FORWARDATT(o, U)
  map U to dimension of o to get $\hat{U}$
  Calculate c by Eq. (5)
  Calculate B by Eq. (6)
  Return $B \times U$
end function

To make the pipeline more clear, we would like to describe the whole process by pseudocode. We denote the input image patches by X, the output classification by P, the total number of epochs by n_e, and the output of block s by $U(s)$. A detailed pseudocode can be found in Algorithm 1.

IV. EXPERIMENTS

Our experiments are carried out on three popular HSI data sets: the Pavia University data set, the Indian Pines data set, and the Salinas Scene data set. We evaluate the performance of the proposed attention-aware P3D CNN for HSI classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

TABLE I
TRAINING AND TEST SAMPLE SIZES USED IN THE PAVIA UNIVERSITY DATA SET

Class No.	Class Name	Training	Test
1	Asphalt	548	6631
2	Meadows	540	18649
3	Gravel	392	2099
4	Trees	524	3064
5	Metal Sheets	265	1345
6	Bare Soil	532	5029
7	Bitumen	375	1330
8	Bricks	514	3682
9	Shadows	231	947
	Total	3921	42776

TABLE II
TRAINING AND TEST SAMPLE SIZES USED IN THE INDIAN PINES DATA SET

Class No.	Class Name	Training	Test
1	Alfalfa	50	1384
2	Corn-notill	50	784
3	Corn-min	50	184
4	Corn	50	447
5	Grass-pasture	50	697
6	Grass-trees	50	439
7	Grass-pasture-mowed	50	918
8	Hay-windrowed	50	2418
9	Oats	50	564
10	Soybean-notill	50	162
11	Soybean-mintill	50	1244
12	Soybean-clean	50	330
13	Wheat	50	45
14	Wood	15	39
15	Buildings-grass-trees	15	11
16	Stone-steel-tower	15	5
	Total	695	9671

The input image will be first divided into $8 \times 8 \times$ bands patches. After that, the extracted patches will pass through the neural network, and finally, the output classification map is obtained. More detailed settings and results are described in the following sections.

A. Experimental Data Set Descriptions

We test on three data sets, whose descriptions are as follows.

The Pavia University scene was acquired by the ROSIS sensor during a flight campaign over Pavia, Northern Italy. This data set consists of nine classes, whose number of spectral bands is 103 after removing 12 noise channels. Pavia University is with 610×340 pixels, but some samples in the image contain no information and thus are discarded before the analysis. The geometric resolution is 1.3 m. The numbers of training and test samples are given in Table I.

The Indian Pines was gathered by an airborne visible/infrared imaging spectrometer sensor over the Indian Pines agricultural site in Northwestern Indiana in June 1992. It presents 16 classes, mostly related to land covers. This data set consists of 145×145 pixels with a spatial resolution of 20 m per pixel and 10-nm spectral resolution over the range of 400–2500 nm. In this article, after removing 20 bands affected by atmosphere absorption, we made use of 200 bands. The numbers of training and test samples are given in Table II.

The Salinas Scene data set was acquired by the 224-band AVIRIS sensor over Salinas Valley, CA, USA. This data set has a spatial resolution of 3.7 m, with 204 spectral bands. The image consists of 512×217 pixels, including 16 classes. After preprocessing (radiometric correction, attitude processing, GPS processing, and so on), the final geocorrected image cube is obtained, representing the sensor spectral radiance. The detailed information about the training and testing data is given in Table III.

TABLE III
TRAINING AND TEST SAMPLE SIZES USED IN THE SALINAS DATA SET

Class No.	Class Name	Training	Test
1	Brocoli-green-weeds-1	50	1959
2	Brocoli-green-weeds-2	50	3676
3	Fallow	50	1926
4	Fallow-rough-plow	50	1344
5	Fallow-smooth	50	2628
6	Stubble	50	3909
7	Celery	50	3529
8	Grapes-untrained	50	11221
9	Soil-vinyard-develop	50	6153
10	Corn-senesced-green-weeds	50	3228
11	Lettuce-romaine-4wk	50	1018
12	Lettuce-romaine-5wk 1	50	1877
13	Lettuce-romaine-6wk	50	866
14	Lettuce-romaine-7wk	50	1020
15	Vinyard-untrained	50	7218
16	Vinyard-vertical-trellis	50	1757
	Total	800	53329

B. Comparative Methods and Evaluation Metrics

To have a fair and comprehensive comparison, we compare three types of methods. The first type refers to the most popular, traditional machine learning classification methods, including the random forest, support vector machine (SVM), and canonical correlation forest (CCF). The second type refers to deep learning-based methods, including the CNN and the C3D. The third type refers to our proposed methods, including the P3D method and the attention-aware P3D method. More descriptions of the comparative methods are as follows.

RF-200 [47]: It represents the random forest with 200 trees.

SVM-RBF [47]: It means the RBF kerneled SVM method. It is implemented using the libsvm package. Fivefold cross validation is considered to tune the hyperplane parameters.

CCF [47]: Currently, it is probably the best nondeep learning method with the best performance for HSI data. This method constructs a decision forest that consists of multiple canonical correlation trees using canonical correlation components for hyperplane splitting [48].

CNN [13]: In this study, the architecture of CNN includes four convolutional layers, three pooling layers, one fully connected layer, as well as one softmax layer. This layer setting is the same as in the proposed method, as described in Table IV. The only difference is that the 3-D convolution layers are replaced by 2-D convolution layers.

C3D [7]: In this study, the architecture of CNN includes four 3-D convolutional layers, three pooling layers, one fully connected layer, as well as one softmax layer. This layer setting is the same as in the proposed method, as shown in Table IV.

P3D [9]: The only difference between P3D and the C3D is that the C3D blocks in the neural network are replaced by P3D blocks. The model setting can be found in Table IV.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
SETTINGS OF CONVOLUTIONAL NETWORK ARCHITECTURES FOR HSI CLASSIFICATION

Layers	Output Size	3D CNN	P-3D	AP-3D(Ours)
Convolutional Block	8×8×8×32	$\{3\times3\times3$ conv, 32, stride 1$\}\times$ 2, batch norm $\times$ 2	$\left\{\begin{array}{c}3\times3\times1conv,32,stride=1\\1\times3\times3conv,32,stride=1\\3\times1\times3conv,32,stride=1\end{array}\right\}\times1$, $\{3\times3\times3conv,32,stride1\}\times$ 1, add $\times$ 1, batch norm $\times$ 2	$\left\{\begin{array}{c}3\times3\times1conv,32,stride=1\\1\times3\times3conv,32,stride=1\\3\times1\times3conv,32,stride=1\end{array}\right\}\times1$, $\{3\times3\times3conv,32,stride1\}\times$ 1, add $\times$ 1, multiply $\times$ 3, batch norm $\times$ 2
Transition Layer	4×4×4×32	$2\times2\times2$ max pool, stride 2		
Convolutional Block	4×4×4×64	$\{3\times3\times3$ conv, 32, stride 1$\}\times$ 2, batch norm $\times$ 2	$\left\{\begin{array}{c}3\times3\times1conv,32,stride=1\\1\times3\times3conv,32,stride=1\\3\times1\times3conv,32,stride=1\end{array}\right\}\times1$, $\{3\times3\times3conv,32,stride1\}\times$ 1, add $\times$ 1, batch norm $\times$ 2	$\left\{\begin{array}{c}3\times3\times1conv,32,stride=1\\1\times3\times3conv,32,stride=1\\3\times1\times3conv,32,stride=1\end{array}\right\}\times1$, $\{3\times3\times3conv,32,stride1\}\times$ 1, add $\times$ 1, batch norm $\times$ 2
Transition Layer	2×2×2×64	$2\times2\times2$ max pool, stride 2		
Transition Layer	1×1×1×64	$2\times2\times2$ global pool, stride 2		
Dense Layer	$1\times1\times1\times1024$	1024 fully connected		
Transition Layer	$1\times1\times1\times1024$			multiply $\times$ 2, add $\times$ 1
Softmax Layer	1×1×1×9	9 fully connected		
Total Layer		9	9	9
Total Params for Pavia University		0.28M	0.28M	1.39M
Total Params for Indian Pines		0.29M	0.29M	1.41M
Total Params for Salinas Scene		0.29M	0.29M	1.41M
FLOPS for Pavia University		1.28M	1.27M	5.62M
FLOPS for Indian Pines		1.26M	1.25M	4.99M
FLOPS for Salinas Scene		1.29M	1.28M	5.85M

AP3D: This proposed model mainly introduces attention to the input feature as well as the output classification, and the P3D blocks are further changed into AP3D blocks. The model setting can be found in Table IV.

To make the proposed approach fairly comparable with other supervised classifiers, we use the standard sets of training and test samples for the data sets.

For the performance evaluation, following the related work [47], we mainly adopt three performance metrics as follows.

Overall Accuracy (OA): It is the percentage of correctly classified samples against the whole number of samples.

Average Accuracy (AA): It is the average of classification accuracies.

Kappa Coefficient: It is widely used to summarize the accuracy assessment results when evaluating land-use or land-cover classifications in remote sensing. It is generated by the confusion matrix and used for measuring the compatibility between the estimated labels and the ground-truth map.

Confusion Matrix: The column and row of these matrices represent the label of samples. Each element in this matrix means the percentage of samples from row classes being classified into the column classes. The sum of each column/row should be 1.

The OA pays more attention to the whole classification map, and the AA pays more attention to each class, to avoid the accuracy imbalance for each class. For the kappa coefficient, it can efficiently measure the correlations between classes.

C. Detailed Proposed Model Settings

Some implementation details of the proposed architectures can be found in Table IV. For the Pavia University data set, the input is $8\times8\times103$ patches. We first reshape the input to $8\times8\times8\times13$ to create 13 3-D cubes (we add the first band to the last and construct 104 bands, and these 104 bands can be divided into eight sets, each with 13 elements). For the hyperparameters, we use the nadam optimizer, set the learning rate to be 0.00002, and set the batch size to be 32 with 500 training epochs.

For the Indian Pines data set, the network architecture is the same, as shown in Table IV. The input is $8\times8\times200$ patches. We reshape the input to $8\times8\times8\times25$ to divide the input into 25 3-D cubes. For the hyperparameters, we use the nadam optimizer, set the learning rate to be 0.0001 with 0.04 decay, and set the batch size as 16 with 1500 training epochs.

The Salinas Scene data set is also with the same network architecture, as shown in Table IV. The input is $8\times8\times204$ patches, and we reshape the input to $8\times8\times8\times26$ to divide the input into 16 3-D cubes (we add the first four bands to the last and construct 208 bands). For the hyperparameters, we use the nadam optimizer, set the learning rate to be 0.0001, and set the batch as 16 with 1000 training epochs.

It can be noted in Table IV, compared with C3D, P3D mainly changes the convolutional block. For P3D and AP3D, the differences exist in two places. The first is the added attention for the first P3D layer, adding weights to each separate 2-D convolutional layer, which brings three multiplication operations. The second is the attention added on the last fully connected layer, bringing two multiplication and one addition operations.

As noted in Table IV, the total parameters for C3D and the proposed P3D are almost the same, whereas the proposed AP3D has the largest number of parameters because

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

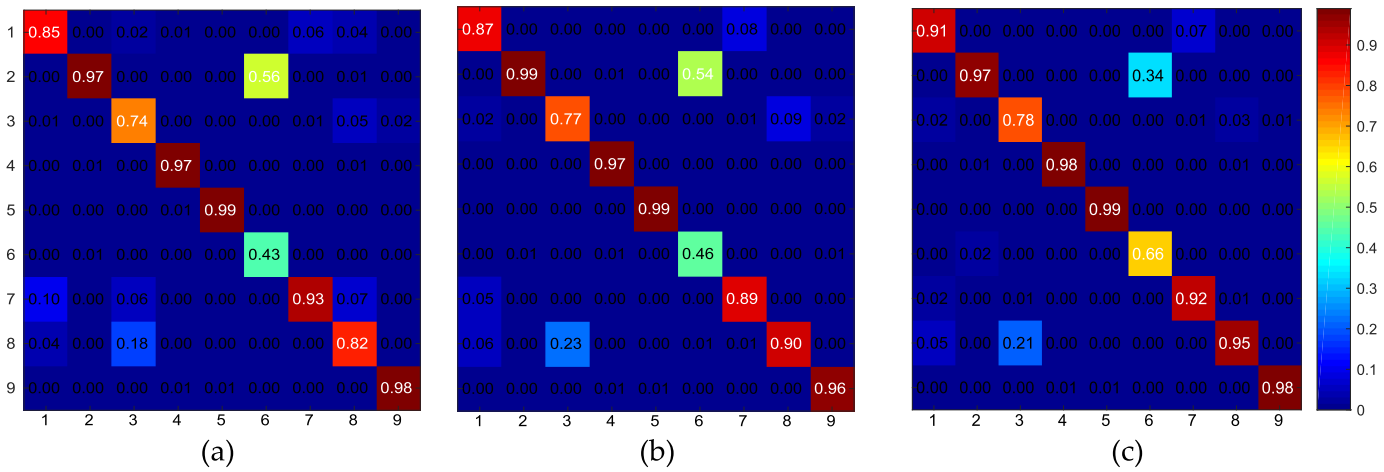


Fig. 3. Confusion matrix comparison for Pavia University among (a) C3D, (b) P3D, and (c) AP3D models.

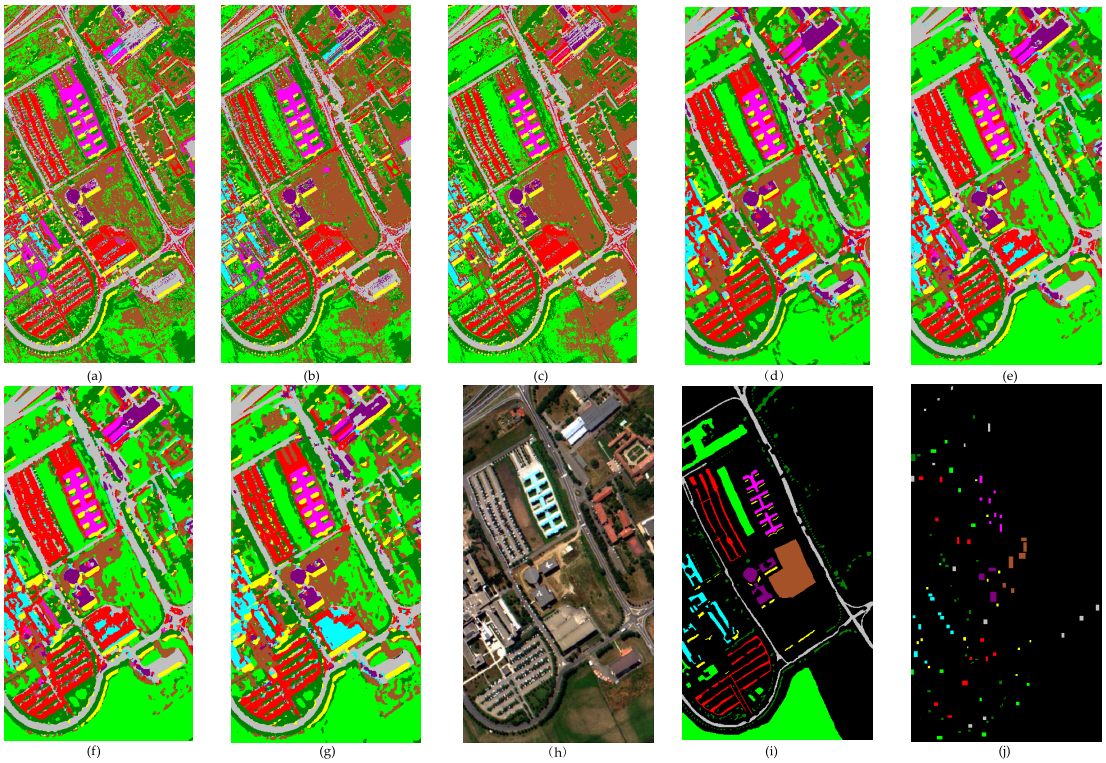


Fig. 4. Classification maps for Pavia University by different models. (a) RF. (b) SVM. (c) CCF. (d) CNN. (e) C3D. (f) P3D. (g) AP3D. (h) True-color composite of the hyperspectral data. (i) Ground truth. (j) Training data.

of the introduction of the attention-aware model. However, these three networks are still with the same level of time complexities.

D. Experiment 1: Pavia University

The first experiment is conducted on the Pavia University data set. For this data set, classification results when employing different methods are reported in Table V, and the confusion matrix of different methods is shown in Fig. 3. The visualized classification map of the proposed methods can be found in Fig. 4.

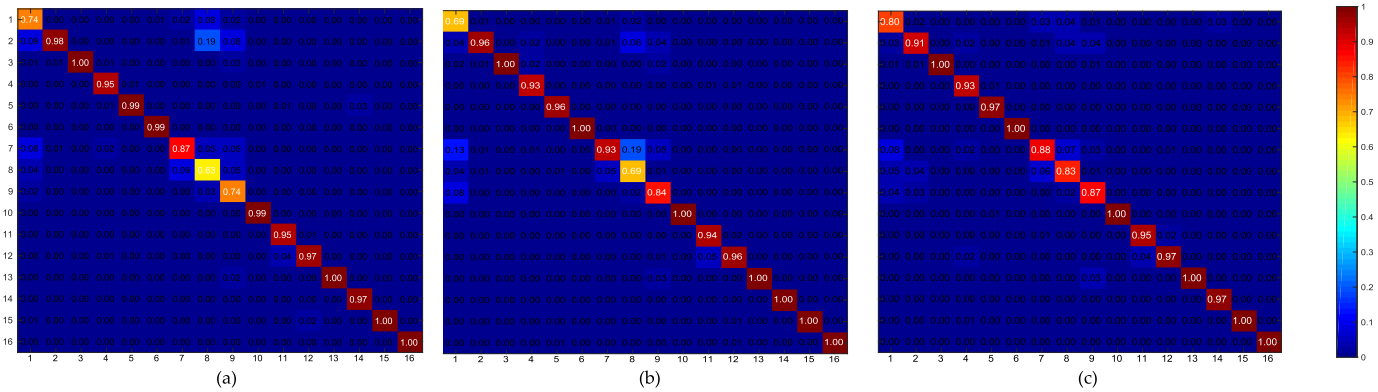
From the table, we can note that the proposed AP3D yields the best performances in terms of OA, AA, and Kappa. Although for some specific categories, the accuracies of AP3D may not be the best, especially for the Bare Soil category. SVM and CCF also provide good results for several classes; however, they fail for the class with the largest amount of samples. This might be due to the overfitting problem. For C3D and P3D, their performances are almost the same (ranked the second), followed by CNN. For RF200, its performance seems the worst.

The performance advantage of 3-D CNN methods is more obvious for the class with the least percentage of training

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

TABLE V
PAVIA UNIVERSITY CASE: CLASSIFICATION PERFORMANCE COMPARISONS

Class No.	Class Name	RF-200	SVM-RBF	CCF-200	CNN	C3D	P3D	AP3D
1	Asphalt	81.54	82.37	86.59	74.48	84.72	86.74	**91.00**
2	Meadows	55.39	67.87	72.33	94.53	97.42	**98.77**	97.13
3	Gravel	53.07	69.18	71.75	64.84	73.80	77.32	**78.23**
4	Trees	98.76	98.37	**99.09**	97.52	97.39	97.29	98.11
5	Metal Sheets	99.11	99.41	**99.78**	99.26	98.51	98.51	98.96
6	Bare Soil	79.10	93.64	**97.26**	55.28	43.23	45.89	65.58
7	Bitumen	84.36	91.20	91.88	**94.89**	92.78	89.47	91.88
8	Bricks	91.39	92.59	94.92	**95.76**	82.40	89.57	95.22
9	Shadows	97.47	96.94	98.73	96.20	97.68	96.20	**98.42**
OA	-	71.53	79.89	83.36	85.87	86.52	88.36	**91.37**
AA	-	82.24	87.95	90.26	85.86	85.33	86.62	**90.50**
Kappa	-	0.6504	0.7491	0.7905	0.8115	0.8178	0.8422	**0.8845**



(a) (b) (c)

Fig. 5. Confusion matrix comparison for Indian Pines image among (a) C3D, (b) P3D, and (c) AP3D models.

samples. For instance, for the Meadows class, its training sample size is 540, while the total test samples are 18 649. For the first three comparison methods, their accuracies are all below 70% whereas for C3D/P3D/AP3D, their accuracies are all above 97%.

Fig. 3 shows the comparisons between the 3-D CNN-based methods. We mainly compare the traditional C3D method with the proposed P3D/AP3D methods. As can be found in the figure, diagonal values of the confusion matrix are much higher for AP3D than for the other two methods. This confirms the effectiveness of the proposed AP3D framework.

In Fig. 4, the whole Pavia University image is classified by all seven models, including the pixels without ground-truth labels. Also, we give out the true-color composite, ground truth, and the training data map.

E. Experiment 2: Indian Pines

The second experiment is conducted on the Indian Pine data set. For this data set, classification results are reported in Table VI, and the confusion matrix of different methods is shown in Fig. 5. The visualized classification maps of the proposed methods can be found in Fig. 6.

For the overall performances, deep learning methods provide the best OA, AA, and Kappa. For the other comparison methods, the OA and AA are around 75% with Kappa around 0.75, whereas for the deep learning methods, the OA and AA are all above 80% with Kappa around 0.85. The proposed AP3D yields the best performance, with 89.14% OA and 94.36% AA. Its performance is much better than that of the second-ranked method.

For each class, it can be noted in the table that larger performance improvements of the proposed P3D/AP3D methods are obtained for the categories including Alfalfa, Hay-windrowed, Soybean-clean, and Woods, which are the most difficult to classify. The comparison methods all fail for these four classes, while the 3-D CNN can perform well for such classifications.

For the confusion matrix, the proposed AP3D yields much higher values for the diagonal elements when compared with the C3D/P3D. All diagonal values are higher than 80%.

In Fig. 6, the whole Indian Pines image is classified by all seven models, including the pixels without ground-truth labels. Also, we give out the true-color composite, ground truth, and the training data map.

F. Experiment 3: Salinas Scene

The third experiment is conducted on the Salinas Scene data set. For this data set, the detailed classification results are reported in Table VII, and the confusion matrix of different methods are shown in Fig. 7. The visualized classification maps of the proposed methods can be found in Fig. 8.

For this data set, all methods achieve more than 80% accuracy for OA. Only the C3D and the proposed AP3D achieve higher than 90% accuracy for OA. The proposed AP3D method yields the highest 93.56% OA accuracy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

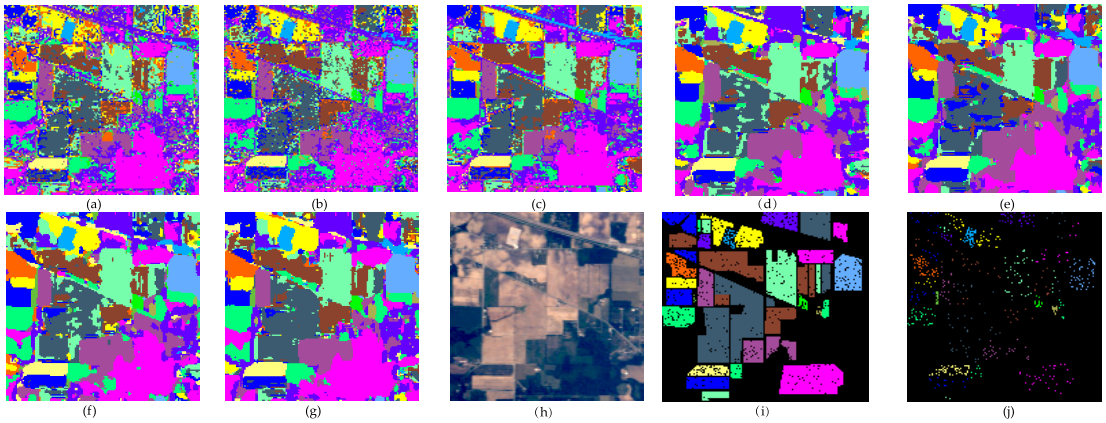


Fig. 6. Classification maps of the Indian Pines image by different models. (a) RF. (b) SVM. (c) CCF. (d) CNN. (e) C3D. (f) P3D. (g) AP3D. (h) True-color composite of the hyperspectral data. (i) Ground truth. (j) Training data.

TABLE VI
INDIAN PINES CASE: CLASSIFICATION PERFORMANCE COMPARISONS

Class No.	Class Name	RF-200	SVM-RBF	CCF-200	CNN	C3D	P3D	AP3D
1	Alfalfa	56.65	71.39	76.37	74.86	74.13	68.71	**80.20**
2	Corn-notill	55.48	71.05	77.93	94.26	**97.96**	95.54	90.94
3	Corn-min	82.07	86.96	94.57	97.28	100	100	**100**
4	Corn	85.23	91.72	94.41	92.17	**95.08**	93.29	93.29
5	Grass-pasture	80.20	85.80	91.39	**99.71**	98.57	96.27	97.27
6	Grass-trees	94.99	93.85	97.04	99.32	99.09	99.54	**99.77**
7	Grass-mowed	77.02	75.38	90.96	92.27	87.15	**92.81**	88.45
8	Hay-windrowed	57.94	59.88	69.48	73.99	63.19	68.61	**83.29**
9	Oats	62.94	76.24	**89.01**	79.43	73.58	84.40	87.06
10	Soybean-notill	95.06	96.91	98.77	**100**	99.38	**100**	**100**
11	Soybean-mintill	88.67	79.58	93.73	93.01	94.69	94.21	**95.10**
12	Soybean-clean	53.33	74.84	74.55	96.06	96.67	96.06	**96.97**
13	Wheat	97.78	97.78	**100**	**100**	**100**	**100**	**100**
14	Woods	56.41	79.49	97.44	**100**	97.44	**100**	97.44
15	Buildings-grass	81.82	**100**	90.91	**100**	**100**	**100**	**100**
16	Stone-steel	**100**	**100**	**100**	**100**	**100**	**100**	**100**
OA	-	69.31	74.24	82.87	86.00	82.98	84.24	**89.14**
AA	-	76.60	83.80	89.78	93.27	92.31	93.09	**94.36**
Kappa	-	0.6538	0.7093	0.8059	0.8409	0.8075	0.8218	**0.8761**

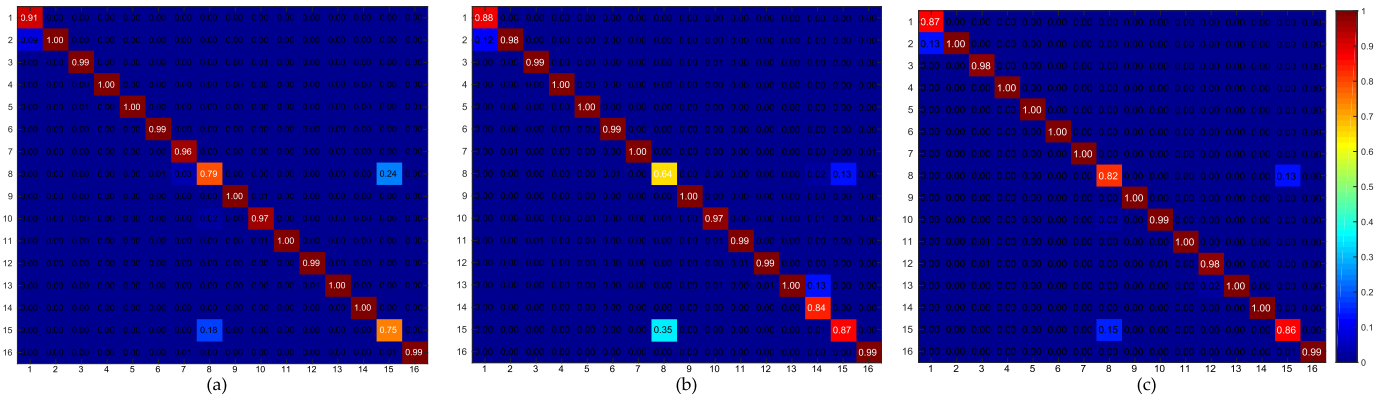


Fig. 7. Confusion matrix comparison for Salinas Scene among (a) C3D, (b) P3D, and (c) AP3D models.

Higher performance improvements of the proposed methods are found in the two largest classes that are difficult to classify. They are Grapes-untrained and Vinyard-untrained. The sample sizes of these two classes are 11 221 and 7218, and the total number of samples for these two takes up almost half of the data. For Grapes-untrained, the best accuracy of the comparison methods is around 70%, while the proposed AP3D model achieves 82.10% accuracy; for the Vinyard-untrained, the deep learning-based methods almost all achieve 85%+ accuracy, whereas the nondeep learning-based methods are all around 75% accuracy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

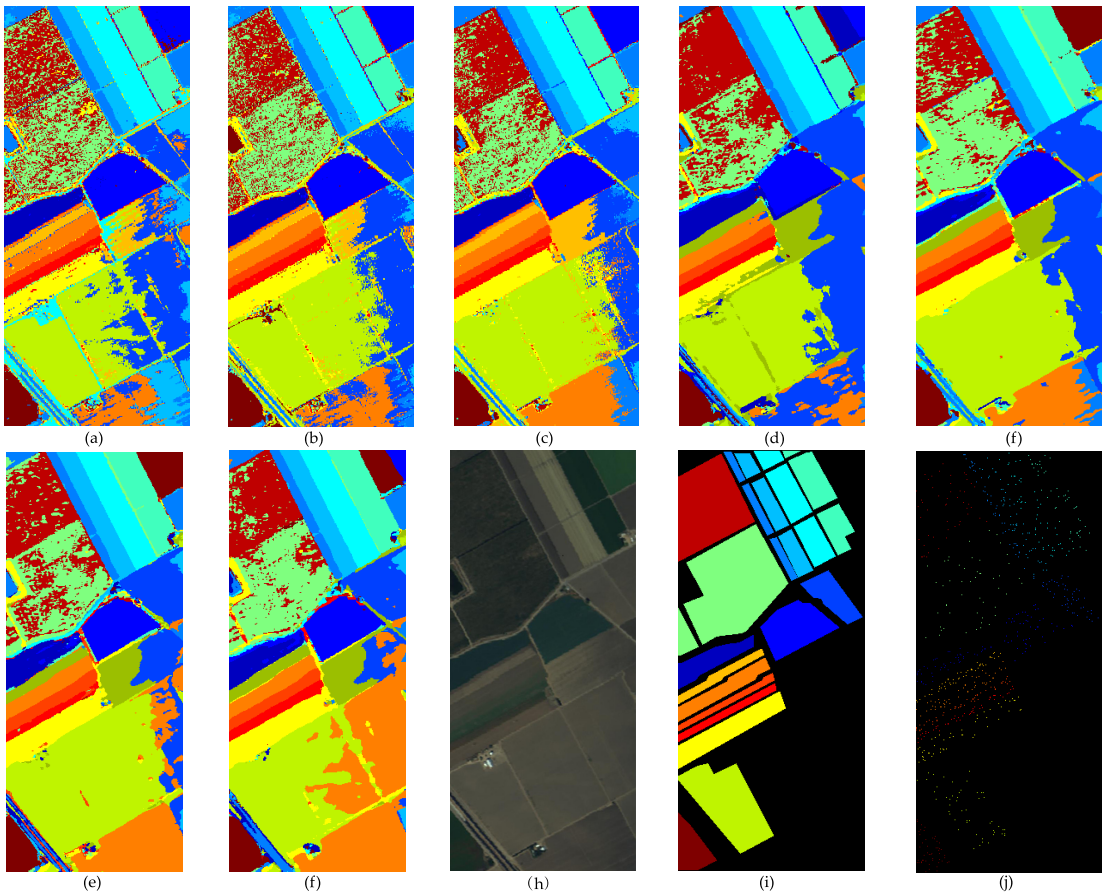


Fig. 8. Classification maps of the Salinas Scene image by different models. (a) RF. (b) SVM. (c) CCF. (d) CNN. (e) P3D. (f) C3D. (g) AP3D. (h) True-color composite of the hyperspectral data. (i) Ground truth. (j) Training data.

TABLE VII

SALINAS SCENE CASE: CLASSIFICATION PERFORMANCE COMPARISONS

Class No.	Class Name	RF-200	SVM-RBF	CCF-200	CNN	C3D	P3D	AP3D
1	Brocoli-green-weeds-1	99.29	98.98	99.49	**99.80**	91.37	87.75	87.09
2	Brocoli-green-weeds-2	99.21	99.67	99.95	99.81	**100**	98.45	99.73
3	Fallow	97.72	98.70	**99.43**	99.17	98.96	98.55	97.82
4	Fallow-rough-plow	97.62	97.77	99.33	99.48	99.70	**99.85**	**99.85**
5	Fallow-smooth	97.95	98.33	98.82	**99.96**	99.92	99.92	99.89
6	Stubble	99.41	99.72	99.80	100	99.33	98.95	**100**
7	Celery	99.23	99.46	99.66	95.64	95.72	99.63	**99.94**
8	Grapes-untrained	61.92	70.37	67.56	57.61	78.73	64.12	**82.10**
9	Soil-vinyard-develop	98.70	98.59	99.19	99.41	**99.97**	99.92	99.84
10	Corn-senesced-green	85.56	93.74	93.80	72.61	96.69	97.34	**98.61**
11	Lettuce-romaine-4wk	91.75	94.70	95.87	99.80	99.61	99.31	**99.90**
12	Lettuce-romaine-5wk 1	98.24	**99.89**	99.95	97.39	98.93	99.25	98.30
13	Lettuce-romaine-6wk	97.69	97.81	98.15	**100**	**100**	**100**	**100**
14	Lettuce-romaine-7wk	92.25	97.35	96.86	**99.90**	99.71	84.12	**99.90**
15	Vinyard-untrained	70.32	71.53	80.77	94.14	74.91	**87.17**	85.94
16	Vinyard-vertical-trellis	96.98	98.18	98.18	**99.37**	98.52	98.63	**99.37**
OA	-	86.02	88.82	89.72	88.09	91.13	89.44	**93.56**
AA	-	92.74	94.67	95.43	94.63	95.75	94.56	**96.77**
Kappa	-	0.8450	0.8757	0.8858	0.8683	0.9012	0.8828	**0.9283**

For the rest classes, all methods provide more than 90% accuracy. It suggests that they are relatively easy for classification, and the advantages of the proposed methods are not that obvious.

For the confusion matrix, the AP3D yields much higher values for the diagonal elements when compared with the other two methods. There is no value lower than 80%.

In Fig. 8, the whole Salinas Scene image is classified by three models, including the pixels without ground-truth labels. Also, we give out the true-color composite, ground truth, and the training data map.

V. CONCLUSION

In this article, we propose a novel deep learning framework, named attention-aware P3D CNN, for HSI classification.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

The proposed network has the following advantages: it can provide a more detailed description for each dimension of the inputs by assigning attention, and it can better exploit the middle layers' information by considering their compatibility with the final output. When tested on the three most popular data sets, Pavia University, Indian Pines, and Salinas Scene, the proposed method shows great effectiveness and potential for HSI classification.

One potential limit of the proposed method is the model's size, which is slightly larger than that of other 3-D CNN models, although it is still at the same complexity level. For future work, we will explore the network reduction direction for our proposed AP3D. Also, we plan to extend the proposed AP3D model to other data types, such as medical images and natural RGB images.

REFERENCES

[1] M. Zhang, W. Guo, Y. Cui, F. Shen, and Y. Shen, "Manifold learning based supervised hyperspectral data classification method using class encoding," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 3160–3163.

[2] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[3] J. M. Haut, M. E. Paoletti, J. Plaza, J. Li, and A. Plaza, "Active learning with convolutional neural networks for hyperspectral image classification using a new Bayesian approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6440–6461, Nov. 2018.

[4] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "Deep learning classifiers for hyperspectral imaging: A review," *ISPRS J. Photogramm. Remote Sens.*, vol. 158, pp. 279–317, Dec. 2019.

[5] S. Zhao, W. Li, Q. Du, and Q. Ran, "Hyperspectral classification based on Siamese neural network using spectral-spatial feature," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2567–2570.

[6] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Feb. 2019.

[7] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 1–7.

[8] S. Zhao, W. Li, Q. Du, and Q. Ran, "Hyperspectral classification based on Siamese neural network using spectral–spatial feature," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2087–2090.

[9] Z. Qiu, T. Yao, and T. Mei, "Learning spatio-temporal representation with pseudo-3D residual networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 5533–5541.

[10] J. Wu, H. Hu, and L. Yang, "Pseudo-3D attention transfer network with content-aware strategy for image captioning," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 15, no. 3, pp. 1–19, Sep. 2019.

[11] J. Wei, H. Wang, Y. Yi, Q. Li, and D. Huang, "P3D-CTN: Pseudo-3D convolutional tube network for spatio-temporal action detection in videos," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 300–304.

[12] S. Jetley, N. Lord, N. Lee, and P. Torr, "Learn to pay attention," in *Proc. Int. Conf. Represent. Learn.*, 2018.

[13] X. Cao, F. Zhou, L. Xu, D. Meng, Z. Xu, and J. Paisley, "Hyperspectral image classification with Markov random fields and a convolutional neural network," *IEEE Trans. Image Process.*, vol. 27, no. 5, pp. 2354–2367, May 2018.

[14] M. Zhang, W. Li, and Q. Du, "Diverse region-based CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2623–2634, Jun. 2018.

[15] L. Windrim, R. Ramakrishnan, A. Melkumyan, and R. J. Murphy, "A physics-based deep learning approach to shadow invariant representations of hyperspectral images," *IEEE Trans. Image Process.*, vol. 27, no. 2, pp. 665–677, Feb. 2018.

[16] L. Jiao, M. Liang, H. Chen, S. Yang, H. Liu, and X. Cao, "Deep fully convolutional network-based spatial distribution prediction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5585–5599, Oct. 2017.

[17] G. Cheng, Z. Li, J. Han, X. Yao, and L. Guo, "Exploring hierarchical convolutional features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6712–6722, Nov. 2018.

[18] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jan. 2015.

[19] H. Liang and Q. Li, "Hyperspectral imagery classification using sparse representations of convolutional neural network features," *Remote Sens.*, vol. 8, no. 2, p. 99, Jan. 2016.

[20] B. Du, Y. Zhang, L. Zhang, and D. Tao, "Beyond the sparsity-based target detector: A hybrid sparsity and statistics-based detector for hyperspectral images," *IEEE Trans. Image Process.*, vol. 25, no. 11, pp. 5345–5357, Nov. 2016.

[21] S. Mei, J. Ji, Q. Bi, J. Hou, Q. Du, and W. Li, "Integrating spectral and spatial information into deep convolutional neural networks for hyperspectral classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2016, pp. 5067–5070.

[22] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[23] S. K. Roy, S. Chatterjee, S. Bhattacharyya, B. B. Chaudhuri, and J. Platos, "Lightweight spectral–spatial squeeze-and-excitation residual bag-of-features learning for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5277–5290, Aug. 2020.

[24] J. Xu, J. E. Fowler, and L. Xiao, "Hypergraph-regularized low-rank subspace clustering using superpixels for unsupervised spatial-spectral hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, early access, Apr. 29, 2020, doi: 10.1109/LGRS.2020.2985981.

[25] G. Sun *et al.*, "Deep fusion of localized spectral features and multi-scale spatial features for effective classification of hyperspectral images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 91, Sep. 2020, Art. no. 102157.

[26] B. Cui, J. Cui, S. Hao, N. Guo, and Y. Lu, "Spectral-spatial hyperspectral image classification based on superpixel and multi-classifier fusion," *Int. J. Remote Sens.*, vol. 41, no. 16, pp. 6157–6182, Aug. 2020.

[27] B. Tu, C. Zhou, X. Liao, G. Zhang, and Y. Peng, "Spectral–spatial hyperspectral classification via structural-kernel collaborative representation," *IEEE Geosci. Remote Sens. Lett.*, early access, May 29, 2020, doi: 10.1109/LGRS.2020.2988124.

[28] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[29] Y. Li, H. Zhang, and Q. Shen, "Spectral–spatial classification of hyperspectral imagery with 3D convolutional neural network," *Remote Sens.*, vol. 9, no. 1, p. 67, Jan. 2017.

[30] M. He, B. Li, and H. Chen, "Multi-scale 3D deep convolutional neural network for hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Sep. 2017, pp. 3904–3908.

[31] Z. Tian, J. Ji, S. Mei, J. Hou, S. Wan, and Q. Du, "Hyperspectral classification via spatial context exploration with multi-scale CNN," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2018, pp. 2563–2566.

[32] K. Xu *et al.*, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. Int. Conf. Mach. Learn.*, 2015, pp. 2048–2057.

[33] R. A. Rensink, "The dynamic representation of scenes," *Vis. Cognition*, vol. 7, nos. 1–3, pp. 17–42, 2000.

[34] H. Xu and K. Saenko, "Ask, attend and answer: Exploring question-guided spatial attention for visual question answering," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 451–466.

[35] R. Cao, L. Fang, T. Lu, and N. He, "Self-attention-based deep feature fusion for remote sensing scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 43–47, Jan. 2021.

[36] A. Sha, B. Wang, X. Wu, and L. Zhang, "Semisupervised classification for hyperspectral images using graph attention networks," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 157–161, Jan. 2021.

[37] K. Simonyan, A. Vedaldi, and A. Zisserman, "Deep inside convolutional networks: Visualising image classification models and saliency maps," 2013, *arXiv:1312.6034*. [Online]. Available: http://arxiv.org/abs/1312.6034

[38] A. Newell, K. Yang, and J. Deng, "Deep inside convolutional networks: Visualising image classification models and saliency maps," in *Proc. Eur. Conf. Comput. Vis.*, 2016, pp. 1–8.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.

[39] F. Wang *et al.*, "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1–8.

[40] Q. Liu, Z. Li, S. Shuai, and Q. Sun, "Spectral group attention networks for hyperspectral image classification with spectral separability analysis," *Infr. Phys. Technol.*, vol. 108, Aug. 2020, Art. no. 103340.

[41] J. Mun, M. Cho, and B. Han, "Text-guided attention model for image captioning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 31, no. 1, 2017.

[42] J. Li *et al.*, "Hyperspectral image super-resolution by band attention through adversarial learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4304–4318, Jun. 2020.

[43] J. M. Haut, M. E. Paoletti, J. Plaza, A. Plaza, and J. Li, "Visual attention-driven hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 8065–8080, Oct. 2019.

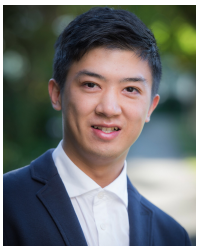
[44] R. Li, S. Zheng, C. Duan, Y. Yang, and X. Wang, "Classification of hyperspectral image based on double-branch dual-attention mechanism network," *Remote Sens.*, vol. 12, no. 3, p. 582, Feb. 2020.

[45] P. H. Seo, Z. Lin, S. Cohen, X. Shen, and B. Han, "Progressive attention networks for visual attribute prediction," 2016, *arXiv:1606.02393*. [Online]. Available: http://arxiv.org/abs/1606.02393

[46] B. Fang, Y. Li, H. Zhang, and J. Chan, "Hyperspectral images classification based on dense convolutional networks with spectral-wise attention mechanism," *Remote Sens.*, vol. 11, no. 2, p. 159, Jan. 2019.

[47] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3755, Jul. 2017.

[48] T. Rainforth and F. Wood, "Canonical correlation forest," Cornell Univ. Liberary, Ithaca, NY, USA, Tech. Rep., 2015.



Jianzhe Lin (Student Member, IEEE) received the Ph.D. degree from the Department of Electrical and Computer Engineering, The University of British Columbia (UBC), Vancouver, BC, Canada, in 2020.

He has been a Post-Doctor with MultiMedia Lab, UBC, since May 2020. He has authored more than 20 papers in top journals and conferences, including IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CYBERNETICS, *Neurocomputing*. His research interests include computer vision, pattern recognition, and machine learning, especially their applications in video surveillance and remote sensing.

Dr. Lin is a regular reviewer for more than 20 distinguished journals. He is/was the Guest Editor of several international journals, such as IEEE GEOSCIENCE AND REMOTE SENSING LETTER, and *Remote Sensing MDPI*.



Lichao Mou (Student Member, IEEE) received the Ph.D. degree from the Technical University of Munich (TUM), Munich, Germany, in 2020.

In 2015, he spent six months at the Department of Computer Science, University of Freiburg, Freiburg, Germany. In 2019, he was a Visiting Researcher with the University of Cambridge, Cambridge, U.K. Since 2019, he has been an AI Consultant with Helmholtz Artificial Intelligence Cooperation Unit (HAICU). He is currently a Guest Professor at the Munich AI Future Lab AI4EO, TUM, and the Head of the Visual Learning and Reasoning Team, Department "EO Data Science," Remote Sensing Technology Institute, German Aerospace Center (DLR), Wessling, Germany.



Xiao Xiang Zhu (Senior Member, IEEE) received the Master (M.Sc.) degree, the Doctor of Engineering (Dr.-Ing.) degree, and the "Habilitation" degree in the field of signal processing from the Technical University of Munich (TUM), Munich, Germany, in 2008, 2011, and 2013, respectively.

She was a Guest Scientist or Visiting Professor at the Italian National Research Council (CNR-IREA), Naples, Italy; Fudan University, Shanghai, China; the University of Tokyo, Tokyo, Japan; and University of California, Los Angeles, CA, USA, in 2009, 2014, 2015, and 2016, respectively. She is currently a Professor for Signal Processing in Earth Observation (www.sipeo.bgu.tum.de) at Technical University of Munich (TUM) and the Head of the Department "EO Data Science," Remote Sensing Technology Institute, German Aerospace Center (DLR), Wessling, Germany. Since 2019, she has been a co-coordinator of the Munich Data Science Research School (www.mu-ds.de). Since 2019, she has been heading the Helmholtz Artificial Intelligence Cooperation Unit (HAICU)—Research Field "Aeronautics, Space and Transport." Since May 2020, she has been the Director of the International Future AI Lab "AI4EO—Artificial Intelligence for Earth Observation: Reasoning, Uncertainties, Ethics and Beyond," Munich. Her main research interests are remote sensing and earth observation, signal processing, machine learning, and data science, with a special application focus on global urban mapping.

Dr. Zhu is a member of the young academy (Junge Akademie/Junges Kolleg) at the Berlin-Brandenburg Academy of Sciences and Humanities and the German National Academy of Sciences Leopoldina and the Bavarian Academy of Sciences and Humanities. She is an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING.



Xiangyang Ji (Senior Member, IEEE) received the B.S. degree in materials science and the M.S. degree in computer science from the Harbin Institute of Technology, Harbin, China, in 1999 and 2001, respectively, and the Ph.D. degree in computer science from the Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China.

He joined Tsinghua University, Beijing, China, in 2008, where he is a Professor with the Department of Automation, School of Information Science and Technology. His research interests include signal processing, image/video processing, image/video compression and communication, 3-D representation, and reconstruction.



Z. Jane Wang (Fellow, IEEE) received the B.Sc. degree in electrical engineering from Tsinghua University, Beijing, China, in 1996, and the M.Sc. and Ph.D. degrees in electrical engineering from the University of Connecticut, Storrs, CT, USA, in 2000 and 2002, respectively.

She has been a Research Associate with the Electrical and Computer Engineering Department, University of Maryland, College Park, MD, USA. Since 2004, she has been with the Department Electrical and Computer Engineering, The University of British Columbia, Vancouver, BC, Canada, where she is a Professor. Her research interest includes statistical signal processing theory and applications, with a focus on multimedia security and biomedical signal processing and modeling.

Dr. Wang received the Outstanding Engineering Doctoral Student Award while at the University of Connecticut. She co-received the EURASIP Journal on Applied Signal Processing (JASP) Best Paper Award in 2004 and the IEEE Signal Processing Society Best Paper Award in 2005. She is the Chair and the Founder of the IEEE Signal Processing Chapter at Vancouver. She is serving as an Associate Editor for the IEEE TRANSACTIONS ON SIGNAL PROCESSING, the IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, and the IEEE TRANSACTIONS ON BIOMEDICAL ENGINEERING.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 19:38:44 UTC from IEEE Xplore. Restrictions apply.